

Samuel Lima Braz

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[Portfolio](#), [GitHub](#), [HuggingFace](#)

PROFILE

Computer engineering graduate who trains, evaluates, and serves vision models, then runs them on hardware. Volunteer at Black Bee Drones on indoor GPS-denied flight: ROS 2, visual odometry, SLAM, detection, and PID control. Industry internships on datasets, model comparison, hyperparameter search, and serving (Triton, ONNX, OpenVINO), including PEFT on vision-language models. Looking toward robotics, computer vision, and models that have to work on a moving machine.

ROBOTICS

Robotics Engineer — Black Bee Drones

Apr 2023 to Sep 2026

- Competed in IMAV, CBR, and SAE Eletroquad. 3rd place indoor at IMAV 2023 (stacking challenge) and IMAV 2025. Special Achievement Award at IMAV 2023 for highly automated MAV operation during the stacking challenge. 2nd place at SAE Eletroquad 2026.
- Lead developer of [Nectar SDK](#), a ROS 2 kit that became the shared software for the team's missions. One flight interface for ArduPilot and PX4 (MAVROS, MAVLink, uXRCE-DDS), plus Bebop and Crazyflie; one camera factory (RealSense, OAK-D, USB, ROS topics); detection, segmentation, and classification (YOLO, DETR, RF-DETR) with training and evaluation. Docker images for x86_64 and ARM64. [Docs](#)
- Indoor navigation without GPS: VIO with Intel RealSense T265 on a Raspberry Pi, then vSLAM with Isaac ROS and RealSense D435i on a Jetson Orin Nano, feeding ArduPilot EKF3 and PX4 EKF2. Wrote a ROS 2 C++ bridge ([vision_to_mavros](#)) from T265 pose into MAVROS, including ENU/NED conversion and covariance.
- PID position control in body, world, and takeoff frames; GPS waypoint navigation with EGM96 geoid correction; obstacle handling from a depth camera. Same missions in Gazebo SITL (ArduPilot and PX4, indoor and outdoor) before flight. Debugged localization and control from flight logs in PlotJuggler. Configurable PID as a ROS 2 node in C++ ([pid-controller](#)).
- OpenCV on the vehicle: HSV/LAB color filters, line estimates (Hough, RANSAC, rotated rectangle, ellipse), ArUco, and distance from a regression on detections. Trained YOLO, DETR, and RF-DETR for competition tasks, converted with TensorRT, OpenVINO, and ONNX, and compared NMS, soft-NMS, WBF, and slicing inference. Integrated Pixhawk with LiDAR, GPS, cameras, motors, and ESCs.

Teaching Member — Fundação Asimo, UNIFEI

Apr 2023 to Aug 2024

Taught programming and embedded systems (Arduino, C++, ESP32, Lego Mindstorms). Built a 4-servo arm with MediaPipe and OpenCV. Mentored students for the Brazilian Robotics Olympiad (OBR). [GitHub](#)

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern — Chunkr

Jul 2025 to Jul 2026

- Built a synthetic document-layout generator from statistics of real pages (positions, co-occurrence, size). Element pool in SQLite with vectorized search and multiprocessing. [Article](#)
- Pretrained YOLO and RF-DETR on 500K+ synthetic samples mixed with real data, then fine-tuned. Stratified rare classes, multi-GPU training, TensorBoard.
- After training: per-class confidence filters, statistical anomaly checks, and box refinement from pixel density. Compared WBF, soft-NMS, and DIOU-NMS and kept a merge that corrected class confusions, raising F1 on every class.
- Fine-tuned LayoutLMv3-Large for reading order: 88.1% exact match and 98.2% Kendall's τ on the open [benchmark](#) (733 samples, 16 document categories). Served models on NVIDIA Triton; workers on Redis.

ML Engineer — Tech4Humans

Feb 2025 to Jul 2025

- Open-source signature detector on Hugging Face (50M+ downloads). [Blog](#) compares YOLO (v8-v12), DETR, and YOLOs; chose YOLOv8s. Public code, weights, data, and report ([collection](#)).
- Optuna search: +7.94 percentage points F1. Triton with ONNX/OpenVINO, <200 ms on CPU.
- Fine-tuned open VLMs ($\leq 12B$) for fields on Brazilian documents with LoRA and QLoRA (Unsloth, MS-Swift), evaluated with vLLM, and served JSON extraction on GPU. Experiment tracking in Weights & Biases and MLFlow.

AI Developer Intern — Tech4Humans

Jul 2024 to Feb 2025

Studied PEFT (LoRA, QLoRA, IA3) for language models under a small compute budget. On dialogue summarization, training less than 1% of parameters matched full fine-tuning closely enough to be useful. Wrote up the methods on the [Hugging Face Community Blog](#) and [AI News Brazil](#).

PROJECTS

Quantum Assistant — Graduation project: synthetic multimodal data and PEFT on Qwen3-VL-8B for Qiskit (circuit diagrams, Bloch spheres, histograms). 8,366-sample dataset; +11–17 pp on Qiskit HumanEval. Published in *Expert Systems with Applications* (Elsevier). [GitHub](#) | [Article](#)

CafeDL — Deep learning library in Java from scratch: Conv2D, Dense, Dropout, optimizers, ND4J tensors, and a QuickDraw-style classifier in JavaFX. [GitHub](#)

EDUCATION

Bachelor's degree in Computer Engineering, UNIFEI

Mar 2022 to Dec 2025

Technical High School in Industrial Automation, IFMG

Jan 2019 to Feb 2022

Skills: Python, C++, C, Java, Git, Docker, PyTorch, OpenCV, ROS 2, Gazebo, Isaac ROS, Optuna, Triton, ONNX, OpenVINO, TensorRT